Identifying the Best Defensive First Baseman in Baseball

Data Collection, Modeling Plan, and Deployment Steps

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Objective

Build a rigorous, reproducible system to evaluate first-base defense by decomposing it into: (i) Range & Reaction, (ii) Receiving (scoops, stretches, off-target throws), and (iii) Decision-Making (bunts, bag coverage). Convert performance above expectation into runs saved and aggregate with principled uncertainty for player rankings that inform R&D, coaching, and roster decisions for the Philadelphia Phillies.

1 Data Sources & Access

- **Statcast / Baseball Savant**: pitch-by-pitch and batted-ball data (EV, spray, fielder positions when available), play outcomes, runner states. (Public; rate-limited API.)
- **Hawk-Eye/Tracking (not available)**: fine-grained player/ball trajectories, throw vectors to 1B.
- **Video** (MLB.com/YouTube): raw inputs for CV on receiving plays, bunts, and range. Store as MP4 with metadata (gameId, inning, pitchId).
- **Run Expectancy Tables**: RE24/RE288 for converting events to runs saved.

2 Data Collection Plan

2.1 Event Selection

- Plays where the fielder at 1B is involved: putouts at 1B, assists to 1B, errors at 1B, bunts toward 1B, grounders to right side, throws originating from Infield (SS/2B/3B/P) to 1B.
- Include context: count, outs, base/runner states, shift/positioning flag if available, batter/pitcher handedness, park.

2.2 Statcast Pulls

1. Build a daily job to fetch: `game_pk, at_bat_number, pitch_number, batter, pitcher, events, hit_coordinates, hit_speed, hit_angle, fielder_positioning, throw_to_1B` .
2. Store to a normalized architecture: `games`, `pitches`, `balls_in_play`, `fielding_events`, `players`.

2.3 Video Acquisition & Indexing

1. For each selected play, download a short clip (5s pre-contact to 3s post-outcome). Prefer high-FPS center/first-base camera.
2. Name convention: `VID_gamePk_abXX_pitchYY_1B_playerId.mp4`.
3. Create an index (`csv/json`) linking clip to event data.

3 Annotation & Labeling

3.1 Automatic Labels (from tracking/events)

- Primary fielder, throw origin, time-of-flight, hop count, scoop attempt flag, stretch posture flag.
- Batted-ball vector and time-to-arrival at 1B bag zone.

3.2 Human QA

- Use Label Studio/VIA to correct auto labels on a small diversified sample.
- Annotator agreement checks on "scoop vs. non-scoop", "off-target left/right/high/low".

4 Computer Vision Pipeline

4.1 Detection & Tracking

- Detect 1B and ball: YOLOv11 (pretrained → fine-tune on baseball frames).
- Track trajectories: DeepSORT for 1B; Kalman filter for ball with constant-acceleration model.

4.2 Pose Estimation & Key Geometry

- Pose: YOLOv11n-pose to extract shoulder, elbow, wrist, hips, knees, ankles.
- Derived angles: arm extension, lateral stretch angle, trunk bend; glove-to-throw path distance; foot-to-bag vector.

4.3 Throw Reconstruction

- Fit a parametric path to ball detections near catch time; estimate arrival angle, vertical offset at glove plane, hop type.
- Classify throw difficulty (*in-dirt short hop*, *wide left/right*, *high*, *on-line*).

5 Feature Engineering

For each play, compute features grouped by component:

- **Range/Reaction**: time-to-ball, lateral distance to intercept, initial positioning, batter handedness, EV/LA/spray.

- **Receiving**: throw arrival offset (x,y,z), hop amount, 1B stretch angle, glove travel distance, release-to-catch time.
- **Decision**: bunt indicator, runner states, pitcher/batter handedness, game state leverage (LI), historical team coverage rules.

Include park, inning, fatigue (pitches faced), shift flag, and camera/angle controls.

6 Modeling Strategy

6.1 Model A: Range & Reaction (Spatial Catch Probability)

- Target: success probability of fielding the ball, conditional on context.
- Methods: Gradient-boosted trees.
- Output: play-level expected success p_{range} and runs saved using RE change.

6.2 Model B: Receiving (Scoop/Stretch Success)

- Target: probability of successful catch given throw vector & mechanics.
- Methods: Logistic regression with splines (interpretable) + XGBoost (performance baseline).
- Output: p_{rec} and runs saved from converting likely errors into outs.

6.3 Model C: Decision-Making (Policy Evaluation)

- Compare chosen action (field ball vs cover bag vs charge bunt) to league-optimal action by expected run value.
- Methods: Contextual bandit. Inputs: state vector (runners, outs, bunt prob, hitter speed), action taken, outcome.
- Output: decision value above/below league policy in runs.

6.4 Model D: Aggregation & Uncertainty

- Sum play-level runs: $\Delta R = \sum(\text{actual} - \text{expected})$ per component.
- Bayesian hierarchical model to stabilize player estimates (partial pooling by season/team/park).
- Final score: Defensive Runs Saved at 1B with credible intervals.

7 Validation & Calibration

- Split by *game* and *season* to avoid leakage; cluster-robust standard errors by player.
- Report log loss, calibration curves for probability models.
- Year-to-year reliability, out-of-sample ranking stability.
- Face validity checks vs OAA/DRS; ablation to show CV features add lift.

8 Run-Value Mapping

- Use RE24: $\Delta R = \text{RE(after)} - \text{RE(before)}$ for observed vs expected outcomes.
- Weight high-leverage plays naturally via state RE without manual tuning.

9 Deliverables

9.1 Artifacts

- Reproducible pipeline: `make` or `dvc` stages for data pulls, CV inference, modeling, and reports.
- Dashboard: top 1B rankings with component breakdowns; player cards with video examples.
- Tech report: methodology, validation, and Phillies-focused insights.

9.2 Deployment Hooks

- Nightly batch on latest games; incremental updates to player estimates.
- API/CSV drops for coaching staff; annotated clips for player development.

10 Timeline

	Milestone
1	Statcast data, event filters, sample video set (200 plays)
2	CV baseline: ball/1B detection & tracking; pose on stills
3	Receiving model (labels + logistic), range features prototype
4	Range model; run-expectancy link; validation
5	Decision model; aggregation + Bayesian shrinkage
6	Dashboard + report draft; add Phillies player case studies
7	Scale to full season; robustness checks; polish deliverables

11 Risks & Mitigations

- **Video quality/angles**: diversify sources; focus receiving model.
- **Label noise**: human-in-the-loop quality checks.
- **Positioning**: include shift/initial position proxies.
- **Small samples**: partial pooling; multi-year priors; report uncertainty.

Appendix: Practical Notes

- File layout: `data/`, `video/`, `labels/`, `models/`, `reports/`.
- Script stubs: `pull_statcast.py`, `make_clips.py`, `infer_cv.py`, `fit_receiving.py`, `fit_range.py`, `fit_decision.py`, `aggregate_runs.py`.